tp

Developer Guide - Ledger 67

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1. Acknowledgements

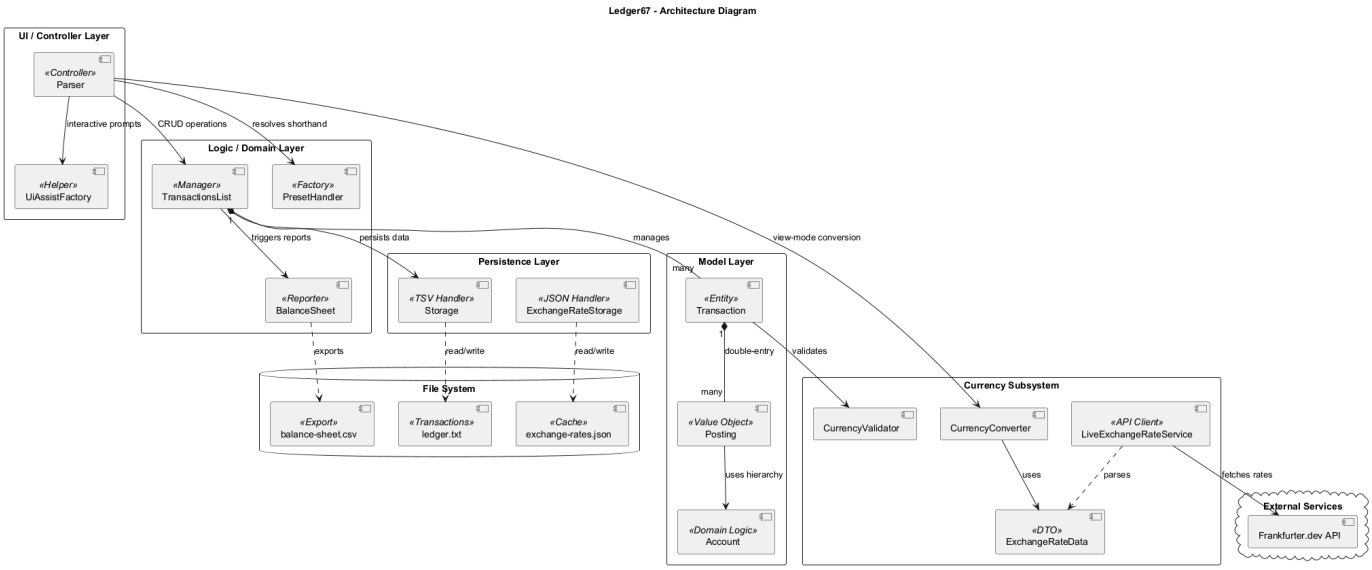
This project is built on the Java platform and follows object-oriented design principles. The application structure is inspired by the individual project iP.

Note: The PlantUML diagram source files are located in the `docs/diagrams/` directory. To generate PNG images from the `.puml` files, use a PlantUML tool or online generator.

2. Design & Implementation

2.1 Architecture Overview

Ledger67 is built on a layered architecture that emphasizes data integrity through double-entry bookkeeping principles. It follows a **Model-View-Controller (MVC)** inspired pattern, adapted for a stateful Command Line Interface (CLI).



1. **Main (Ledger67)**: The entry point. It initializes the infrastructure (Storage, Currency Services), loads existing data, and bootstraps the `Parser` to start the execution loop.

2. **Controller (Parser & UiAssistFactory)**:
    - **Parser**: The central engine that interprets CLI inputs. It features a **Stateful Workflow** for multi-step commands (e.g., `convert` → `confirm`).
    - **UiAssistFactory**: An interactive layer that intercepts commands when "UI Assist" mode is active, prompting users for required fields to build valid CLI arguments.
3. **Logic Layer (TransactionsList & PresetHandler)**:
    - **TransactionsList**: The domain manager. It handles filtering (Regex, Date, Account), CRUD operations, and invokes the `BalanceSheet` engine.
    - **PresetHandler**: A factory that transforms shorthand keywords (e.g., `DAILYEXPENSE`) into full double-entry posting sets.
4. **Model Layer (Transaction, Posting, Account)**: Encapsulates the financial logic, ensuring that every entry adheres to the fundamental accounting equation.
5. **Persistence Layer (Storage & ExchangeRateStorage)**: Handles flat-file persistence (TSV for transactions, JSON for exchange rates).
6. **Currency Engine**: A specialized subsystem consisting of a `LiveExchangeRateService` (API client), `CurrencyConverter`, and `CurrencyValidator`.

2.2 Design Considerations

1. Double-Entry Integrity
Unlike simple expense trackers, Ledger67 enforces a "Balanced" rule for every transaction. A transaction is only committed if the sum of its internal amounts equals zero, following the equation:
Assets = Liabilities + Equity

Where Equity is affected by Income and Expenses: Equity = Initial Equity + (Income - Expenses)

2. Hierarchical Account System
Accounts are not flat strings; they are hierarchical objects (`Account`). This allows users to categorize finances using colons (e.g., `Assets:Bank:DBS`) and enables the system to perform "roll-up" reporting where a filter for `Assets` includes all sub-accounts.

3. Hybrid Input Modes
The system bridges the gap between power users and beginners. It supports a **Flag-based CLI** (fast, regex-compatible) and an **Interactive UI Assist** mode (guided, prompt-based) that generates the underlying CLI commands for the user.

4. Stateful Confirmation Workflow
To prevent accidental data modification during currency conversion or bulk listing, the `Parser` implements a "Pending state." This allows users to preview converted values in "View-Only" mode before explicitly committing them via a `confirm` command.

5. Monetary Precision Ledger67 stores monetary values using **2 decimal places**, which is standard for currency amounts.

If users enter values with more than 2 decimal places, Ledger67 will round them before storing them.

Examples:

- `100.999999` is stored as `101.00`
- `50.1` is stored as `50.10`
- `75.555` is stored as `75.56`

5. Sign Convention

- Ledger67 uses a **signed amount model** instead of debit/credit.
- Ledger67 only checks that transactions are **balanced**, and does not enforce sign conventions.
- Users are encouraged to follow the recommended usage for consistency.

2.3 Component-Level Design

2.3.1 Parser & UI Component

- **Flag Parsing**: Uses regex-based tokenization to support non-positional arguments (e.g., `-date`, `-desc`, `-p`).
- **Command Delegation**: Maps input to specialized handlers like `handleAdd`, `handleList`, and `handleConvert`.
- **UI Assistance**: When enabled, `UiAssistFactory` uses a `Scanner` to interactively collect data, ensuring mandatory flags are never omitted.

2.3.2 Transaction Management Component

- **Filtering Engine**: Utilizes Java Streams and functional predicates to layer filters. Users can filter by account prefix, date ranges, and case-insensitive regex descriptions simultaneously.
- **Balance Sheet Generation**: Aggregates totals across all accounts, calculates “Net Income” from Income/Expense accounts, and validates the accounting equation. It provides dual output: a formatted console view and an automated CSV export for external analysis.

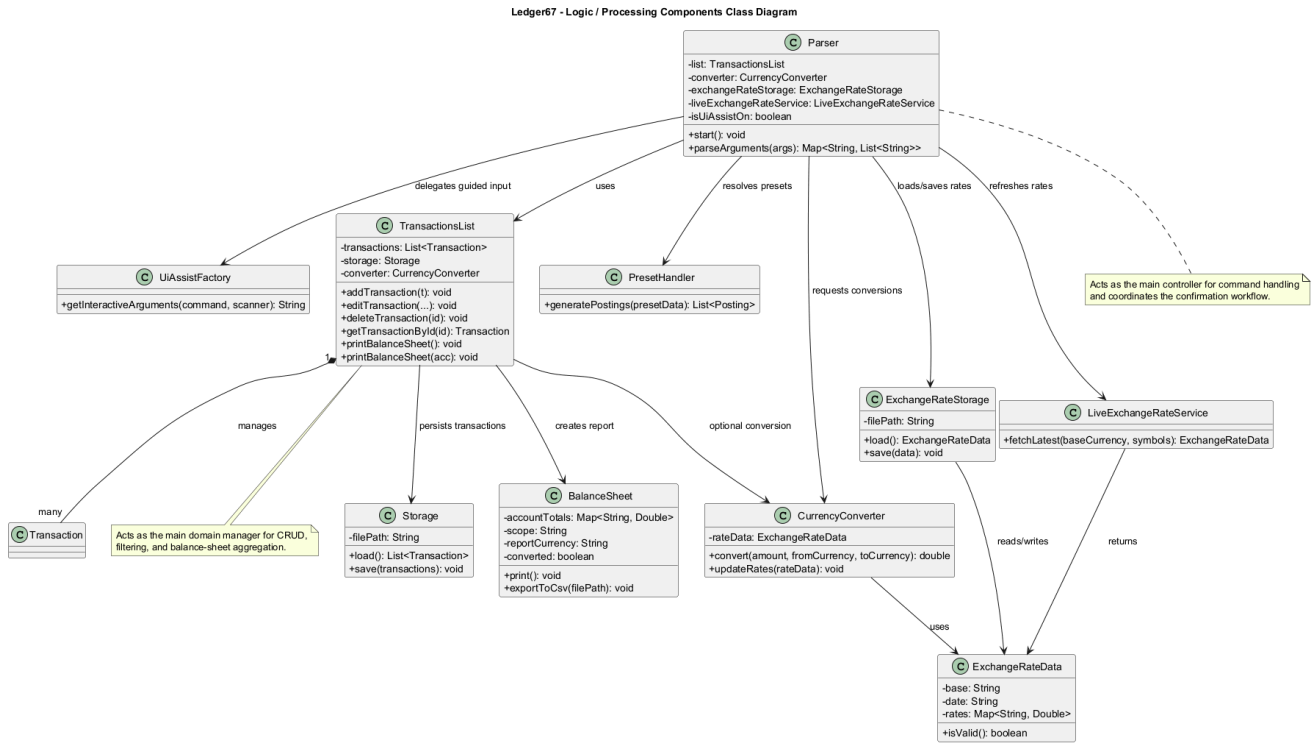
2.3.3 Currency Engine

- **External Integration**: `LiveExchangeRateService` fetches real-time data from the Frankfurter API using Java’s `HttpClient`.
- **Conversion Logic**: The `CurrencyConverter` uses a base-currency (EUR) triangulation method to convert between any two supported currencies (SGD, USD, EUR) based on cached or live

rates.

2.3.4 Class diagram Component-Level Design

- The following class diagram summarizes the main processing components and their dependencies.



2.4 Model Components

- **Transaction**: The core entity. It contains an auto-incrementing ID, metadata (Date, Description, Currency), and a `List<Posting>`. It acts as the final gatekeeper for data integrity via the `isBalanced()` check.
- **Posting**: Represents a single entry in a ledger. It links an `Account` to a `double` amount. It handles the sign-convention logic (e.g., determining if a positive value in an "Income" account represents an increase or decrease in the internal equation).
- **Account**: Manages the validation of "Root" accounts (Assets, Liabilities, Equity, Income, Expenses) and handles the logic for hierarchical membership (`isUnder`).
- **ExchangeRateData**: A Data Transfer Object (DTO) used for GSON serialization/deserialization of currency rates.

2.5 Data Persistence

- **Transaction Storage**: Uses a custom tab-separated format (`ledger.txt`). Complex objects like the list of `Postings` are serialized into a semicolon-delimited string within the file to maintain a flat structure while preserving the multi-entry nature of double-entry bookkeeping.

- **Rate Storage**: Uses `exchange-rates.json` to cache API results, allowing the application to function offline using fallback or previously fetched data.

3. Implementation Details

3.1 Transaction Flow

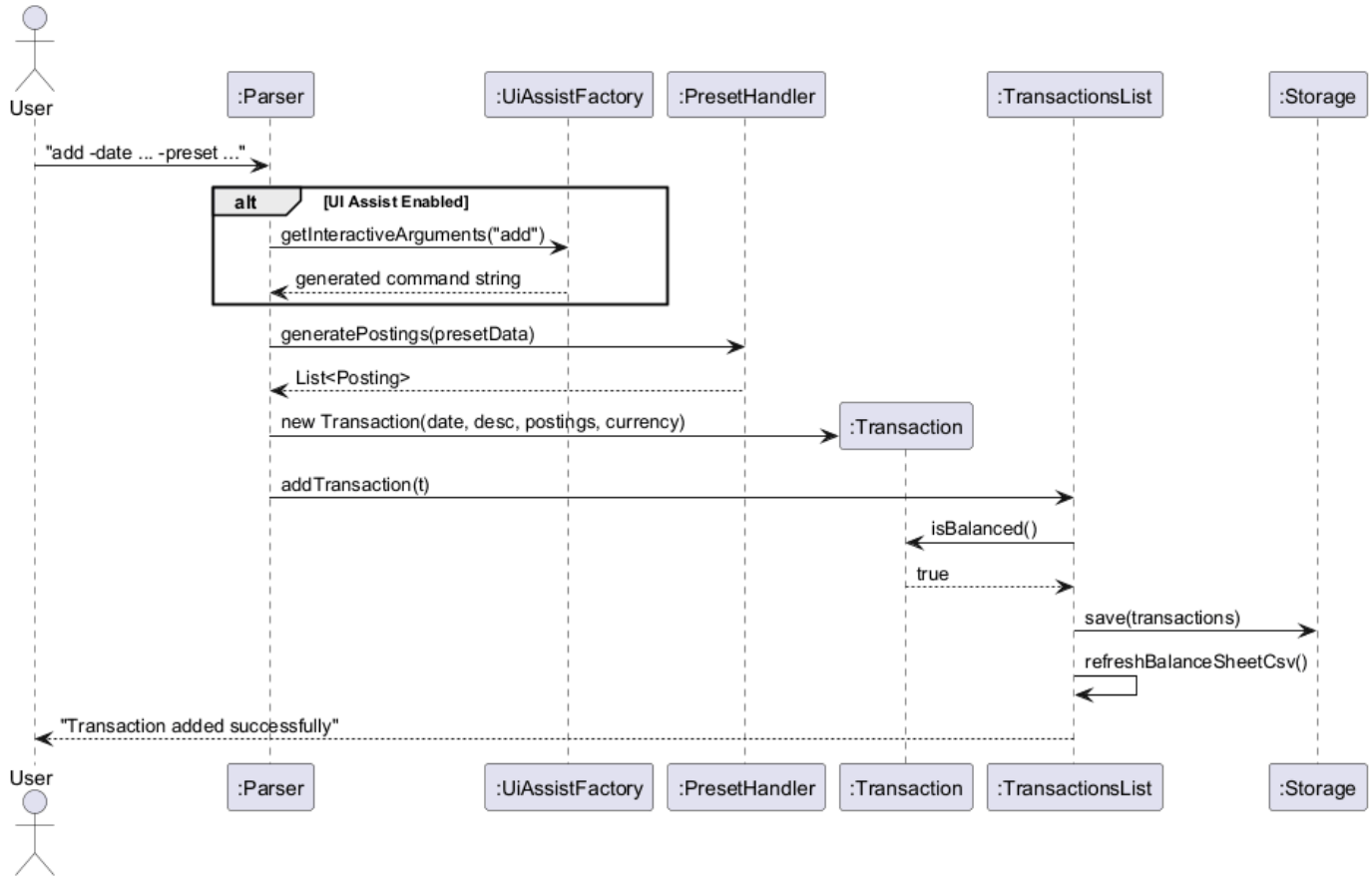
Implementer: Pran, JJ

The transaction flow manages the lifecycle of financial records, ensuring that every entry satisfies the fundamental accounting equation before being persisted.

3.1.1 Create (Add) Flow

The addition process supports two paths: **Manual Entry** (providing specific postings) or **Preset Entry** (using templates).

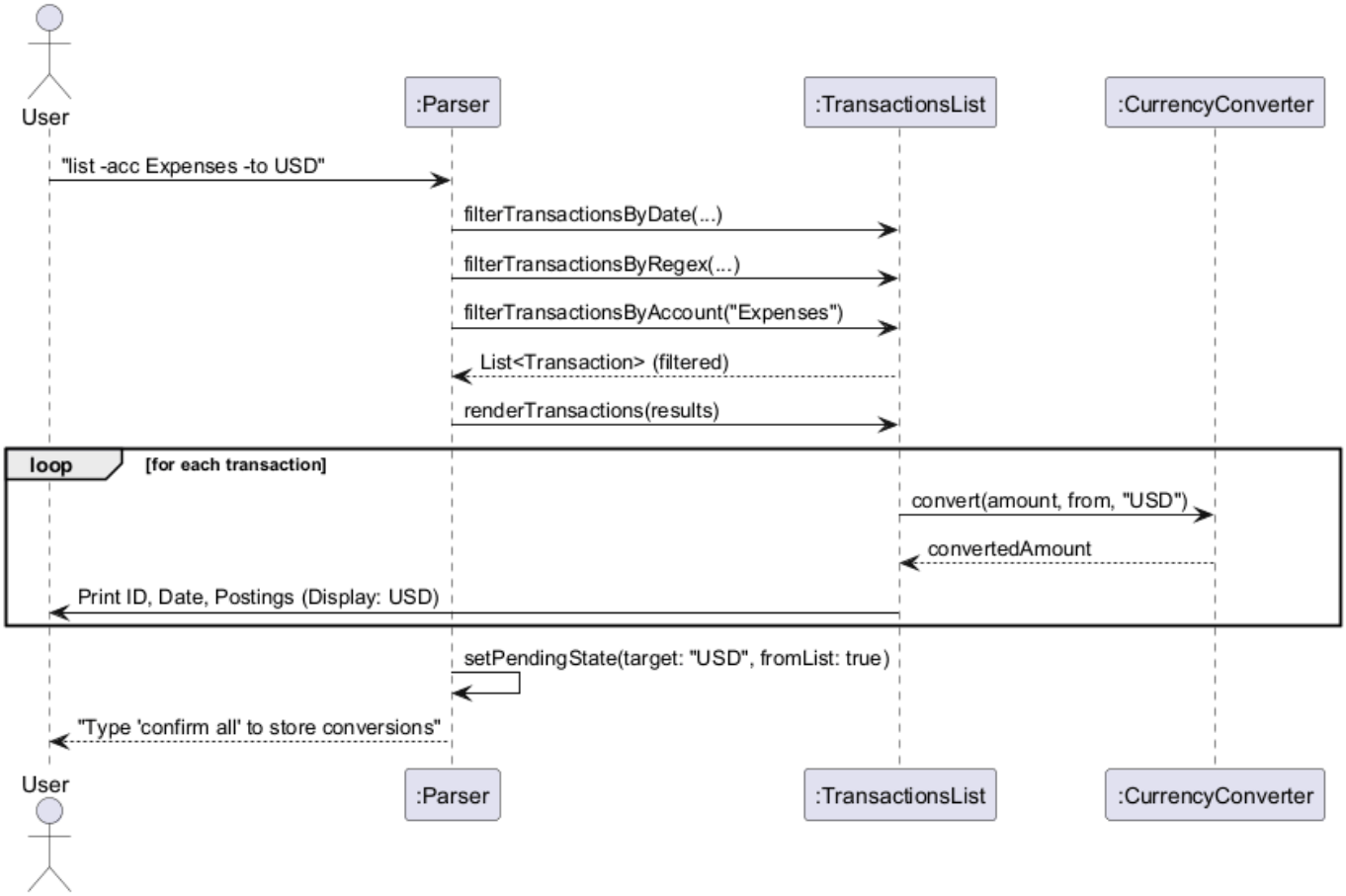
- **Input**: If `isUiAssistOn` is true, the `UiAssistFactory` interactively prompts the user for fields. Otherwise, the `Parser` extracts flags directly.
- **Processing**:
  - If a `-preset` is used, the `PresetHandler` generates a `List<Posting>` (e.g., swapping Assets for Expenses).
  - The `Parser` validates the currency and date format.
  - The `Transaction` object is instantiated and must pass an `isBalanced()` check (Sum of postings ≈ 0).
- **Finalization**: `TransactionsList` appends the transaction, triggers `Storage` to save, and refreshes the background `balance-sheet.csv` .



3.1.2 Read (List/Filter) Flow

Listing is non-destructive and supports layered filtering.

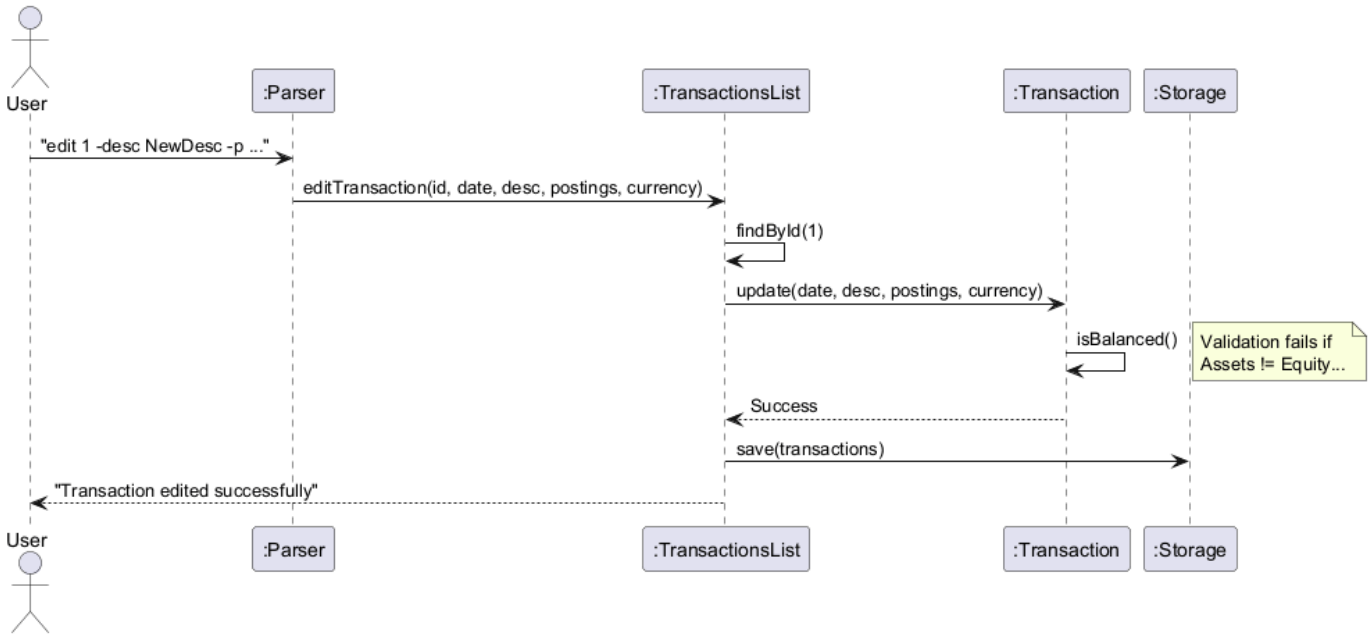
- **Filtering**: `TransactionsList` applies filters in sequence: Date Range → Regex Match → Account Hierarchy (e.g., filtering for `Assets` includes `Assets:Cash`).
- **View-Only Conversion**: If the `-to` flag is present, the `Parser` sets a **Pending State**. The `TransactionsList` renders the converted values for preview but does not modify the underlying data unless a `confirm` command follows.



3.1.3 Update (Edit) Flow

Updates allow modifying any part of an existing transaction while enforcing re-validation.

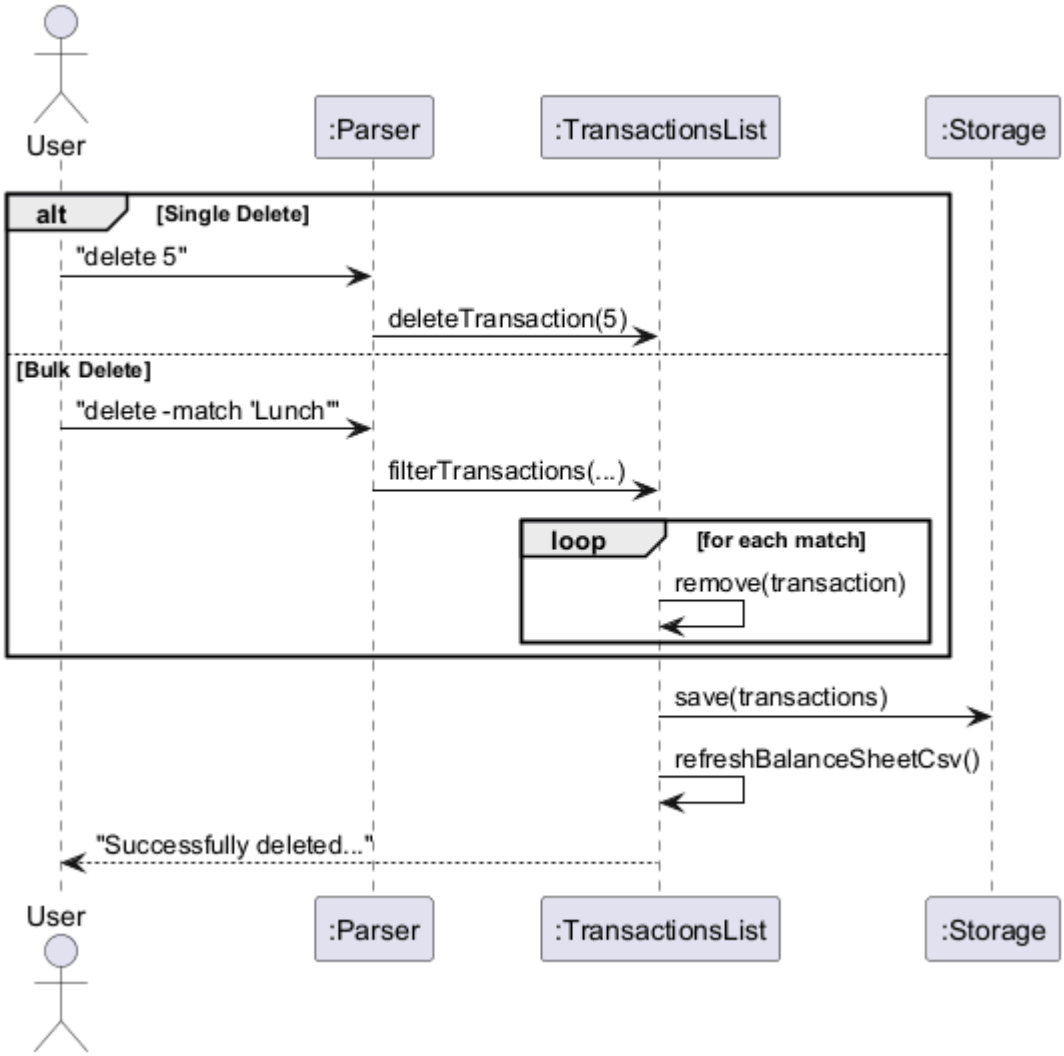
- **Lookup**: `TransactionsList` retrieves the transaction by ID.
- **Modification**: The `Transaction.update()` method selectively replaces fields (Date, Desc, Currency, or Postings).
- **Validation**: If postings are updated, the `isBalanced()` check is re-run. If the new set is unbalanced, the update is aborted.



3.1.4 Delete Flow (Single & Bulk)

Deletion supports targeted removal or mass-clearing based on filters.

- **Single**: Deletes a specific transaction by its unique ID.
- **Bulk**: Uses the same filtering engine as the `list` command to identify a sub-set of transactions for removal.
- **Safety**: Bulk deletion requires at least one filter flag to prevent accidental total wipes (users must use `clear` for that).



3.2 Updated Transaction Data Model

The `Transaction` model is designed for **Double-Entry Bookkeeping**:

- **Identity & Metadata**: Contains a unique `int id`, `LocalDate date`, `String description`, and `String currency`.
- **Postings**: Instead of a single amount, it holds a `List<Posting>`. Each `Posting` links a specific `Account` to a `double` amount.
- **Account Logic**: The `Account` class validates that the root is one of the five standard types: *Assets, Liabilities, Equity, Income, Expenses*. It handles colon-delimited hierarchy (e.g.,

`Expenses:Food`).

- **Balance Integrity**:
  - `Posting.getInternalAmount()` : Translates external user numbers into internal accounting signs based on the account type (e.g., a positive number in `Expenses` is a debit, while a positive number in `Income` is a credit).
  - `Transaction.isBalanced()` : Sums all `internalAmounts` . The transaction is valid only if the total is zero (within an epsilon tolerance of 0.0001 tolerance for floating-point precision).
- **Persistence**: Serialized as a Tab-Separated Value (TSV) line where postings are encoded as `Account1=Amount;Account2=Amount` .

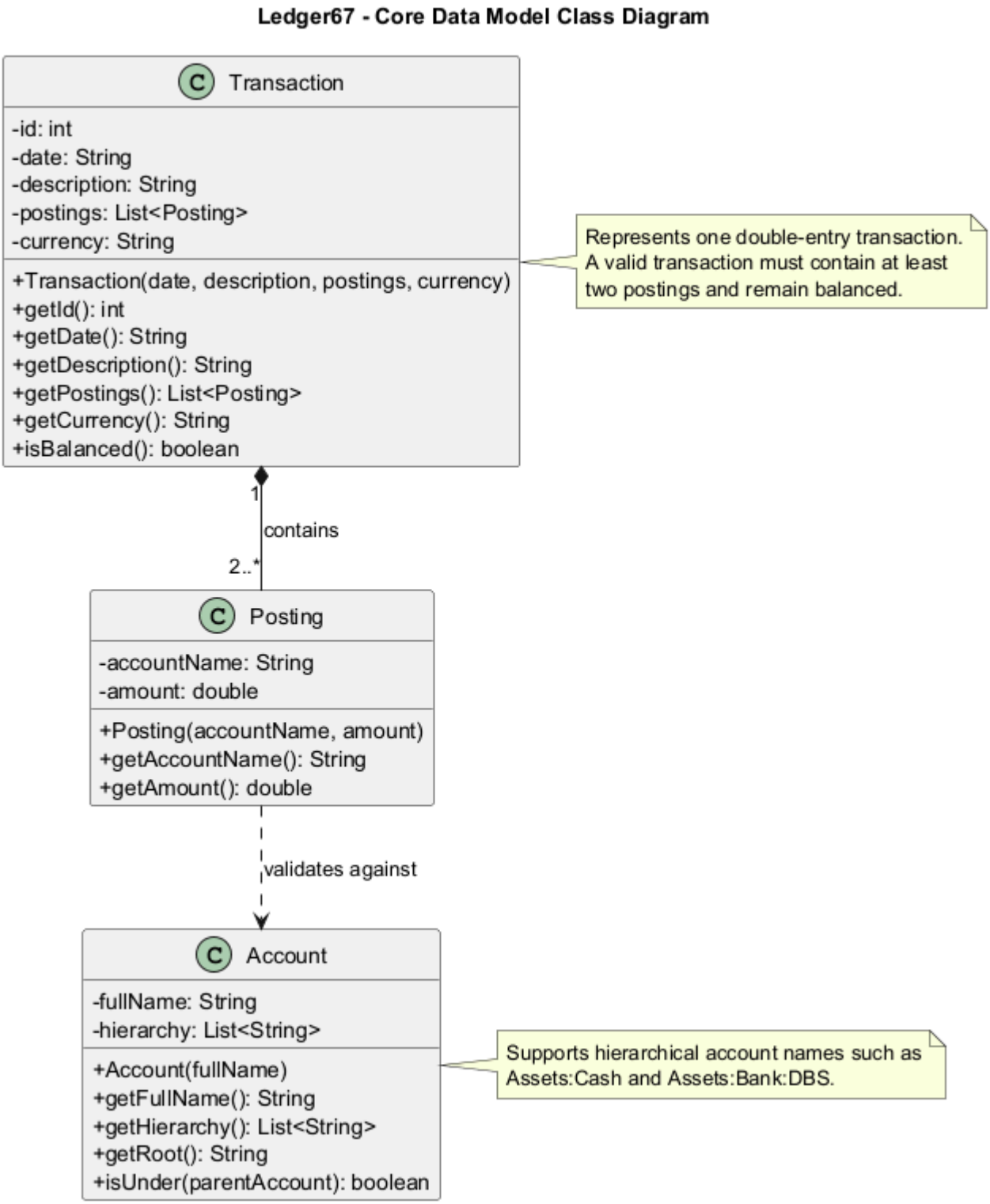
3.2.1 Design Considerations

- **Validation Timing (Balanced vs. Unbalanced):**
  - The system checks if the sum of all postings in a transaction equals zero.
  - **Decision:** We allow the creation of an "unbalanced" transaction object in memory during the parsing phase to provide specific feedback (e.g., "Your transaction is off by 5.00"), but we prevent it from being committed to the `TransactionsList` or `Storage` .
- **Atomicity:**
  - By grouping all postings into a single `Transaction` object, we ensure that an "Office Supply" purchase either fails entirely or succeeds entirely. It is impossible to have an expense recorded without a corresponding deduction from assets.

3.2.2 Alternatives Considered

- **Alternative 1: Flat String Postings:** Storing postings as a simple list of strings within the Transaction class.
  - **Pros:** Easier to parse and store.
  - **Cons:** Extremely difficult to perform mathematical validations or hierarchical filtering.
  - **Decision:** We implemented a dedicated `Posting` class with numeric `amount` and structured `Account` objects to enable rigorous accounting checks.
- **Alternative 2: Single-entry Legacy System:** Storing one account and one amount per line.
  - **Pros:** Simpler CLI syntax (`add -a 50 -acc Food`).
  - **Cons:** Violates fundamental accounting principles and makes it impossible to track where the money came from (e.g., Cash vs. Credit Card).
  - **Decision:** Shifted to the multi-posting `-p` flag system to support true double-entry bookkeeping.

3.2.3 Class Diagram for Core Data Model



3.3 Storage Feature

Implementer: JJ The storage feature is responsible for persisting transaction data to local file storage and restoring it upon application startup.

3.3.1 Transaction Persistence

The `Storage` class manages reading from and writing to a local text file (`ledger.txt`).

- Transactions are saved in a tab-separated format

- Each transaction includes ID, date, description, amount, type, and currency
- Data is written immediately after every modifying operation (add, edit, delete, clear)

This ensures:

- Data durability across application runs
- Minimal risk of data loss

3.3.2 Loading Transactions

Upon application startup, `Storage` loads all previously saved transactions into memory.

- Reads file line-by-line
- Parses each line into a `Transaction` object
- Reconstructs transaction IDs and updates the auto-increment counter
- Invalid or malformed lines are safely skipped to prevent crashes.

3.3.3 Data Encoding and Decoding

To ensure file integrity, special characters are handled using escaping:

- `\t` for tabs
- `\n` for newlines
- `\\` for backslashes This prevents corruption of the file format when storing user input.

3.3.4 Integration with TransactionsList

The `TransactionsList` component interacts directly with `Storage`:

- Calls `load()` during initialization
- Calls `save()` after every modification

This design ensures:

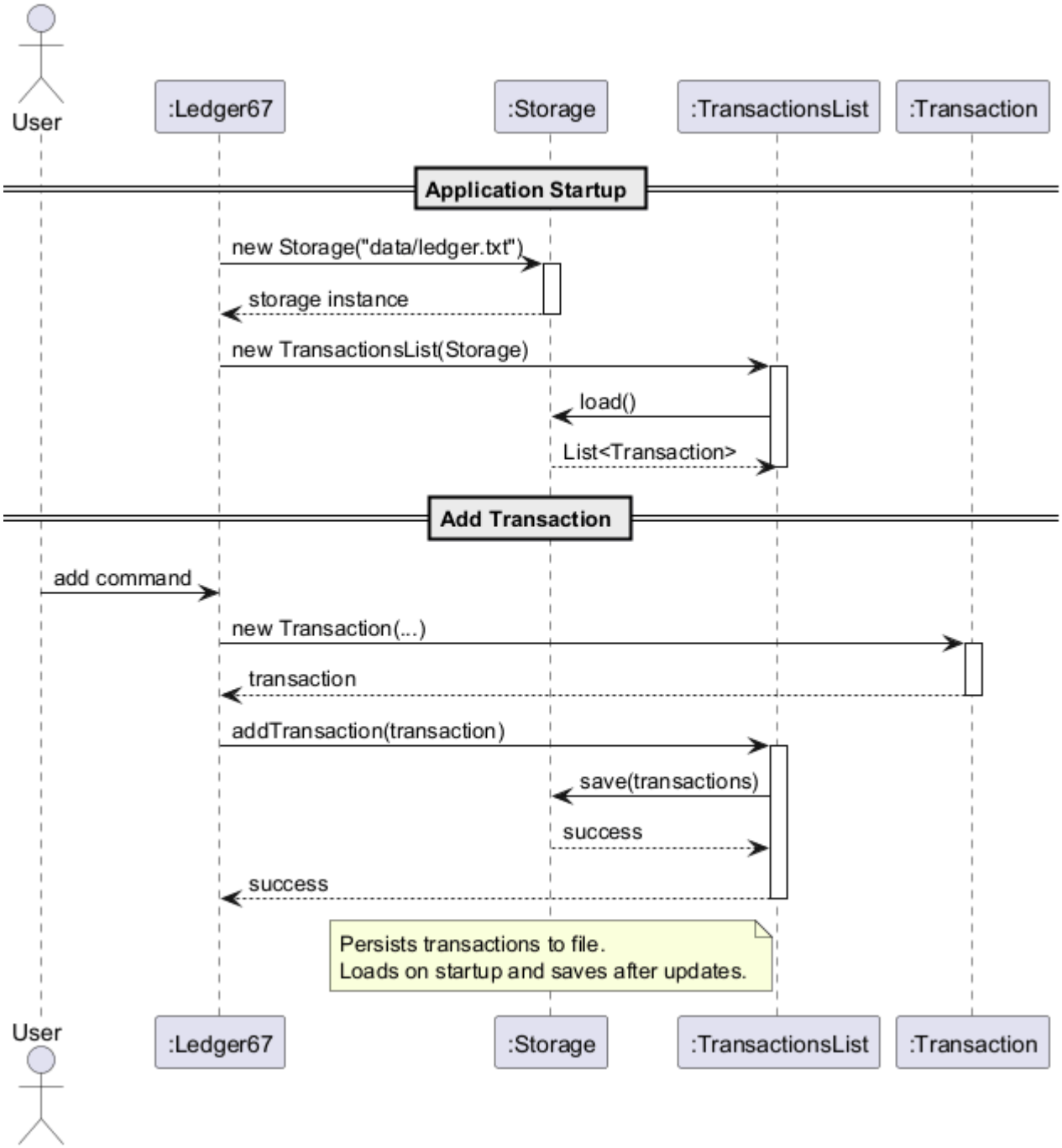
- Separation of concerns between data management and persistence
- Consistent synchronization between memory and disk

3.3.5 Design Considerations

- **Immediate Persistence**: Data is saved after every operation to prevent loss
- **Simple File Format**: Tab-delimited text allows easy debugging and readability

- **Robustness**: Invalid data is safely handled without crashing the application
- **Encapsulation**: Storage logic is isolated from business logic in `TransactionsList`

3.3.6 Sequence Diagram



The diagram above shows:

- `TransactionsList` triggering save operations
- `Storage` writing transaction data to file
- Data being reloaded when the application starts

3.4 Currency Conversion Feature

Implementer: JJ

The currency conversion feature extends the system by introducing dynamic currency handling, persistent exchange rate storage, and real-time rate retrieval.

3.4.1 Integration with Architecture

This feature introduces and integrates the following components:

- `CurrencyConverter` : Core conversion logic
- `ExchangeRateData` : Data model for exchange rates
- `ExchangeRateStorage` : Handles persistent storage of exchange rates
- `LiveExchangeRateService` : Fetches real-time exchange rates

These components are connected as follows:

- `Ledger67` initializes the converter and injects it into `Parser` and `TransactionsList`
- `Parser` handles `convert` and `rates` commands
- `TransactionsList` uses the converter for display-level conversions
- `ExchangeRateStorage` ensures exchange rates persist across application runs

3.4.2 Simple Currency Conversion

The `CurrencyConverter` class provides the core conversion logic via the `convert()` method.

- Validates currencies using `CurrencyValidator`
- Converts via a base currency for consistency
- Handles same-currency conversion as a no-op

3.4.3 Exchange Rate Storage

The `ExchangeRateStorage` component ensures exchange rate data is persisted locally.

- Loads exchange rates from a JSON file at application startup
- Saves updated rates after fetching from the API
- Ensures data validity before reading or writing

This allows the application to:

- Avoid repeated API calls
- Maintain functionality even without internet access

3.4.4 Live Exchange Rate Integration

The `LiveExchangeRateService` fetches real-time exchange rates from an external API.

- Sends HTTP requests to retrieve latest rates
- Parses JSON responses into `ExchangeRateData`
- Updates local storage via `ExchangeRateStorage`
- Triggered using the `rates refresh` command

A fallback mechanism in `Duke` ensures the system remains functional if live data retrieval fails.

3.4.5 Conversion of Stored Transactions

Stored transactions can be converted dynamically without modifying underlying data. Implemented in:

- `Parser` (`convert transaction` command)
- `TransactionsList` (conversion during display)

Features:

- Converts a transaction by ID
- Uses latest available exchange rates
- Preserves original stored values

3.4.6 Display Currency Conversion (List View)

Transactions can be displayed in a selected currency using:

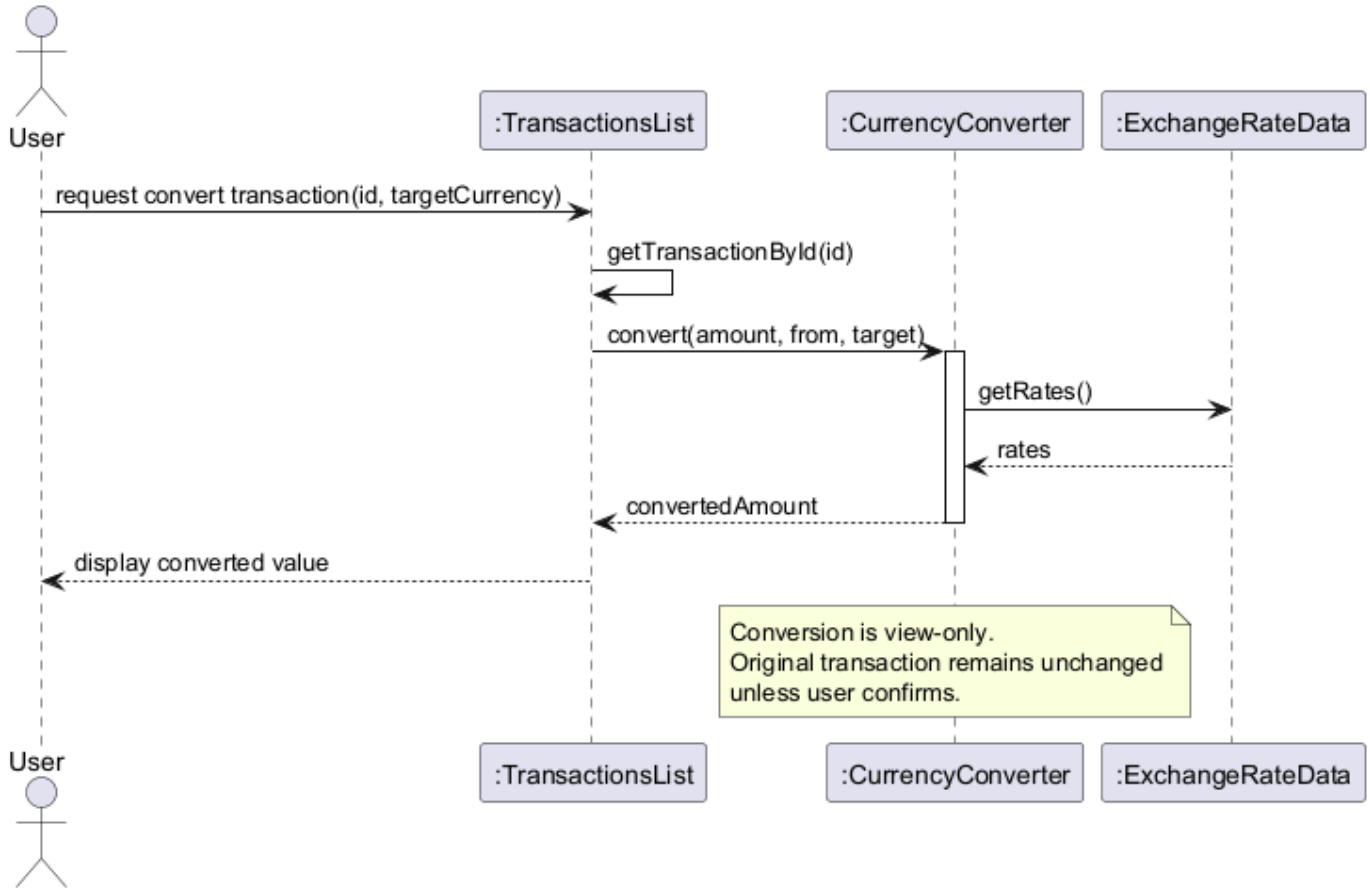
```
list -to USD
```

- Uses `setDisplayCurrency()` and `setAutoConvertDisplay()`
- Conversion is applied during output rendering
- Does not overwrite stored transaction data

3.4.7 Design Considerations

- **Separation of Concerns**: Conversion, storage, and API handling are modularized
- **Persistence**: Exchange rates are stored locally to improve performance and reliability
- **Resilience**: Fallback data ensures continued functionality without API access

3.4.8 Sequence Diagram



The diagram above illustrates how user input flows through the system:

- `Parser` extracts and validates inputs
- `CurrencyValidator` ensures valid currencies
- `CurrencyConverter` performs the conversion
- Results are displayed without modifying stored data

3.5 Confirm and Store Converted Transactions

This feature extends the currency conversion system by allowing users to **persist converted values into storage**, instead of keeping them as view-only. Previously, all conversion operations (`convert` , `convert transaction` , `list transaction -to`) were strictly **non-destructive**.

3.5.1 Implementation Details

This feature is implemented primarily in the `Parser` component through **stateful command handling**.

New internal state variables in `Parser` :

- `pendingTransactionId` : stores the transaction ID (for single conversion)

- `pendingTargetCurrency` : stores the selected target currency
- `pendingFromListView` : indicates whether the conversion came from list view

3.5.2 Workflow

Case 1: `convert transaction`

1. User runs:

   ```
   convert transaction 3 -to SGD
   ```

2. System:
   - Calculates converted value
   - Displays result
   - Stores pending state (ID + currency)
3. User enters:

   ```
   confirm
   ```

4. System:
   - Calls `TransactionsList.editTransaction(...)`
   - Updates:
     - amount
     - currency
   - Persists changes via `Storage.save()`

Case 2: `list -to`

1. User runs:

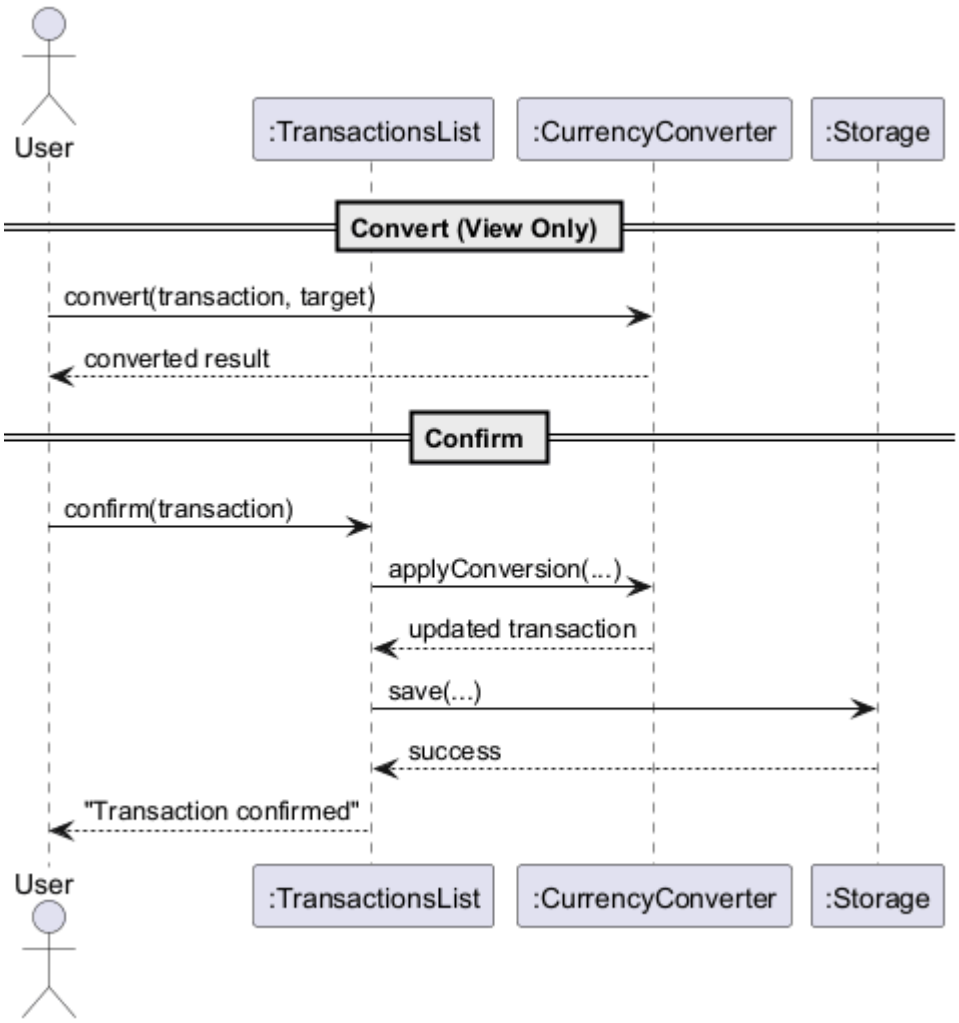
   ```
   list -to SGD
   ```

2. System:
   - Displays converted values (view-only)
   - Stores pending state (currency + list mode)
3. User options:
   - `confirm all` → update all transactions
   - `confirm ID` → update one transaction
4. System:
   - Iterates through transactions (if all)
   - Applies conversion
   - Persists via `Storage`

3.5.3 Design Considerations

- **Explicit Confirmation Required**
  - Prevents accidental data mutation
  - Maintains safety of original financial records
- **Stateful Parser Design**
  - Enables multi-step commands (view → confirm)
  - Avoids modifying core model classes
- **Reuse of Existing Logic**
  - Uses `editTransaction()` instead of creating new update methods

3.5.4 Sequence Diagram



3.6 Transaction Presets Feature

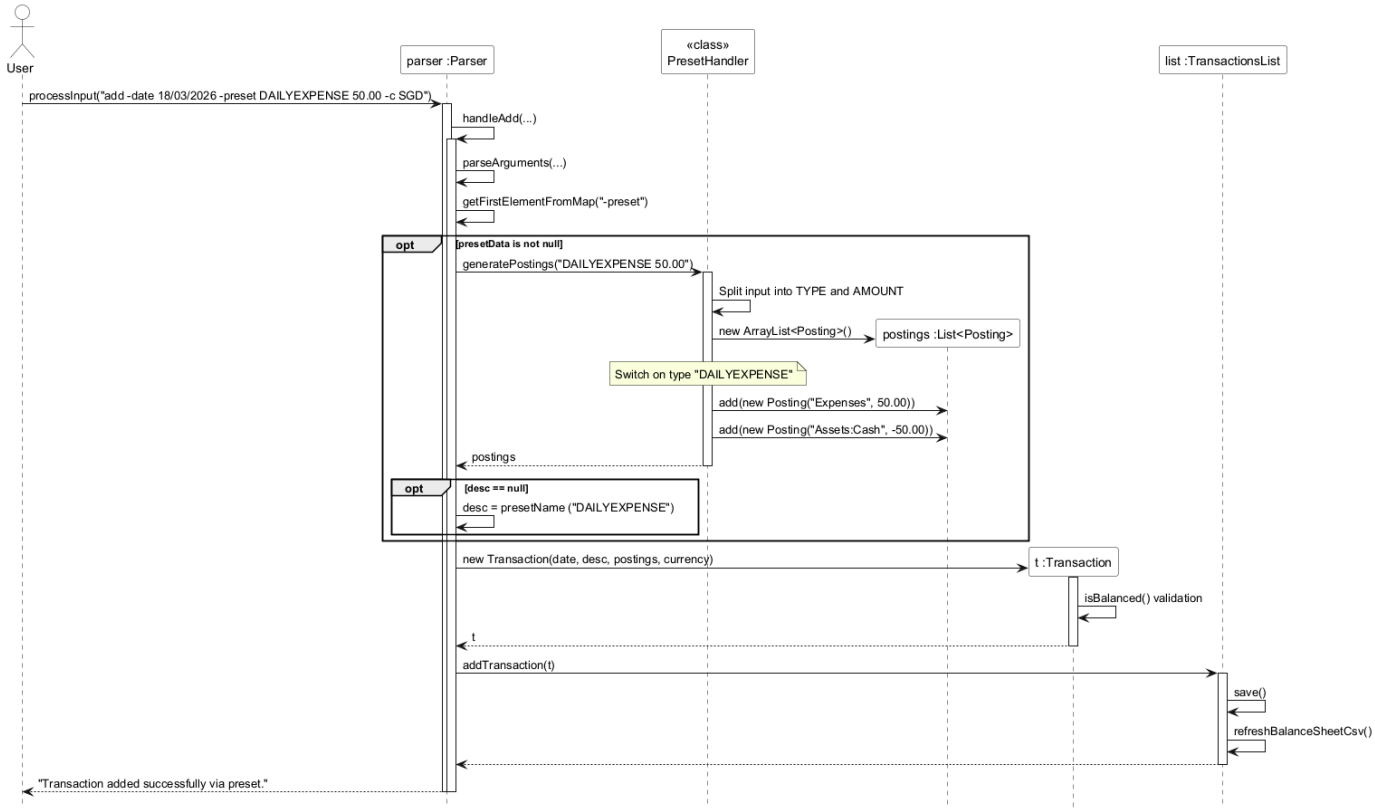
Implementer: Pran

The Preset feature simplifies the user experience by allowing common double-entry transactions to be recorded using a single keyword instead of manual entry for every account posting.

3.6.1 Implementation Details:

The `PresetHandler` class acts as a factory for `Posting` objects.

- **Keyword Mapping**: Keywords like `DAILYEXPENSE`, `INCOME`, and `BUYINGSTOCKS` are mapped to pre-defined accounting movements.
- **Dynamic Amounting**: While the accounts are fixed by the preset, the amount is passed as a parameter during the `add` command.



3.6.1 Workflow:

1. `Parser` detects the `-preset` flag in the `add` command.
2. `Parser` passes the preset string (e.g., `DAILYEXPENSE 50.00`) to `PresetHandler`.
3. `PresetHandler` returns a `List<Posting>` (e.g., `Expenses +50.00`, `Assets:Cash -50.00`).
4. `Transaction` is instantiated using these postings and added to `TransactionsList`.

3.6.2 Design Considerations

- **Defaulting Metadata:**
  - When using a preset, the user provides minimal info.
  - **Decision:** The system automatically injects a default description (e.g., "Daily Expense") if the `-desc` flag is missing, ensuring the data model remains populated while reducing user typing.
- **Extensibility:**
  - The `PresetHandler` uses a Factory pattern.

- **Decision:** This allows developers to add new financial workflows (like `DEPRECIATION` or `LOAN_REPAYMENT`) by simply adding a new case to the factory, without touching the core `Parser` logic.

3.6.3 Alternatives Considered

- **Alternative 1: User-Defined Templates:** Allowing users to save their own custom presets to a file.
  - **Pros:** Highly flexible.
  - **Cons:** Increases complexity of the storage system and requires a "Template Management" UI.
  - **Decision:** Hardcoded presets were chosen for the MVP to provide immediate value for common tasks (Income, Stocks, Expenses) with zero configuration required.

3.7 Balance Sheet Feature

Implementer: JJ

The Balance Sheet feature allows users to generate a real-time summary of their financial position based on the accounting equation:

```
Assets = Liabilities + Equity

Where Equity is affected by Income and Expenses:
Equity = Initial Equity + (Income - Expenses)
```

It aggregates all transactions and computes totals across hierarchical account categories.

3.7.1 Integration with Architecture

This feature primarily extends:

- `Parser` → parses `balance` commands
- `TransactionsList` → computes aggregated values
- `Account` → supports hierarchical filtering
- `CurrencyConverter` → enables optional currency conversion

New component:

- `BalanceSheet` → handles formatting and export logic

3.7.2 Command Handling

The `Parser` handles the following command formats:

```
balance
balance -acc ACCOUNT
balance -to TARGET_CURRENCY
balance -acc ACCOUNT -to TARGET_CURRENCY
```

The parser:

- extracts optional flags (`-acc` , `-to`)
- passes parameters to `TransactionsList.printBalanceSheet(...)`

3.7.3 Balance Aggregation Logic

Implemented in `TransactionsList` :

```
private Map<String, Double> buildBalanceSheetTotals(...)
```

3.7.4 Workflow:

1. Iterate through all transactions
2. Iterate through all postings
3. Apply account filtering:
    - Uses `Account.isUnder(accountPrefix)`
4. Apply currency conversion (if enabled)
5. Aggregate totals into a `Map<String, Double>`

3.7.5 Hierarchical Account Support

The feature leverages:

```
account.isUnder("Assets:Bank")
```

This enables:

- parent-level aggregation (`Assets`)
- sub-account filtering (`Assets:Bank:DBS`)

3.7.6 Income and Expense Handling

Income and Expenses are incorporated into Equity:

```
Net Income = Income - Expenses
```

Implementation:

- Income accounts increase equity
- Expense accounts reduce equity
- Computed dynamically during balance generation

This ensures:

```
Assets = Liabilities + Equity
```

3.7.7 Currency Conversion

If `-to` is specified:

- Each posting is converted using `CurrencyConverter`
- Conversion is applied **during aggregation only**
- Stored data remains unchanged

3.7.8 Output Formatting

Handled by the `BalanceSheet` class:

- Groups accounts into:
  - Assets
  - Liabilities
  - Equity
- Displays:
  - sub-accounts
  - totals
  - accounting check

3.7.9 CSV Export

Each execution of `balance` triggers:

```
balanceSheet.exportToCsv("data/balance-sheet.csv");
```

3.7.10 File Characteristics:

- Overwrites existing file
- Contains:
  - account names
  - balances
  - report currency

3.7.11 Design Considerations

1. Separation of Concerns

- Aggregation → `TransactionsList`
- Presentation → `BalanceSheet`

2. Non-Destructive Operations

- Conversion is view-only
- No mutation of stored transactions

3. Reuse of Existing Structures

- Uses `Posting`, `Account`, `CurrencyConverter`
- Avoids duplicating logic

4. Hierarchical Scalability

- Supports deep account nesting without redesign

3.7.12 Limitations

- Filtered views (`-acc`) may not balance fully
- Mixed currencies require `-to` for meaningful totals
- CSV export is overwrite-only (no versioning)

3.8 Advanced Filtering & Bulk Operations

Implementer: Pran

To manage large datasets, Ledger67 implements a layered filtering system used for both viewing (`list`) and bulk removal (`delete`).

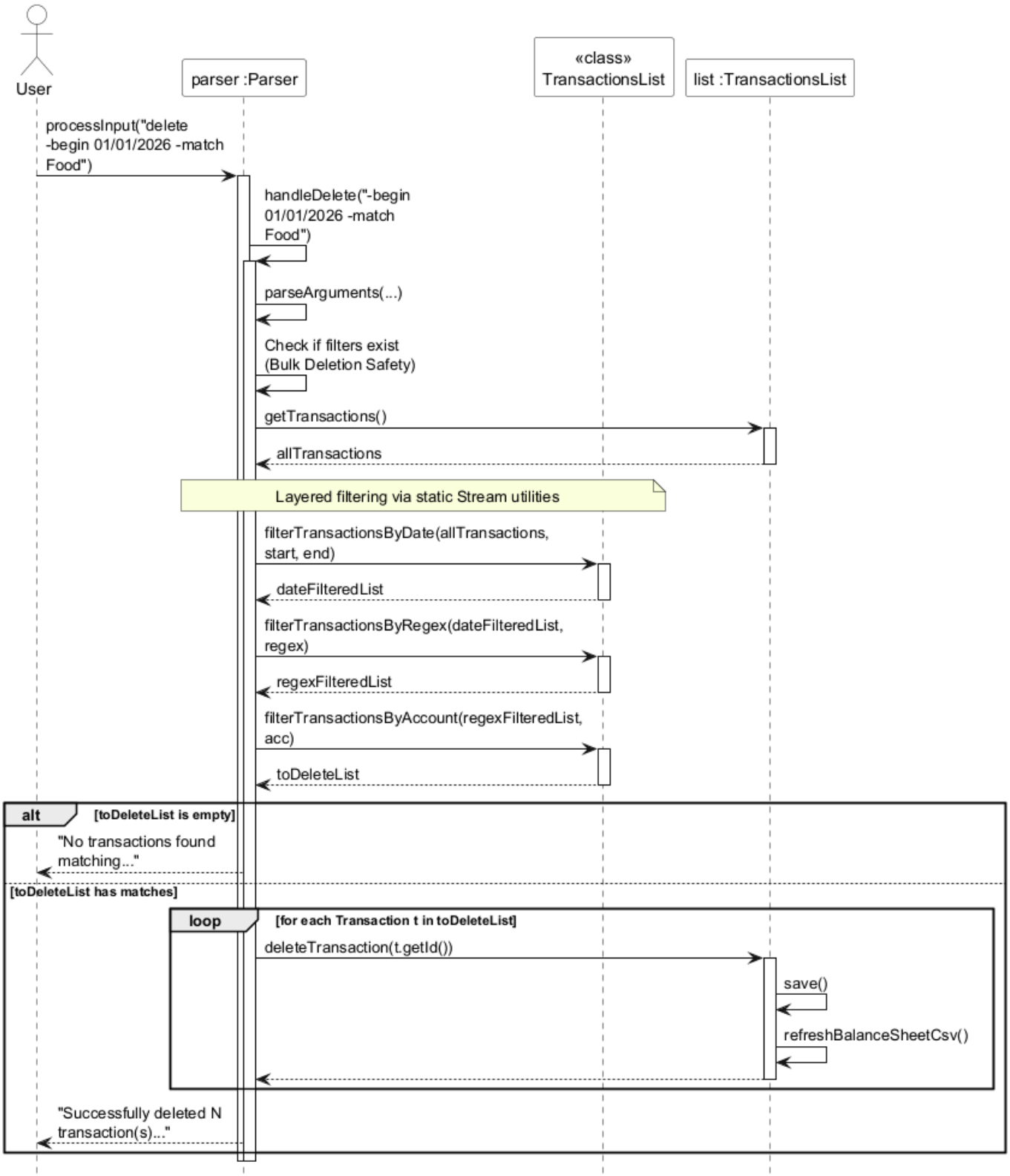
3.8.1 Implementation Details:

Filtering is implemented as a series of static utility methods in `TransactionsList` that utilize Java Streams:

- **Date Filtering**: `filterTransactionsByDate` checks if a transaction falls within an inclusive `LocalDate` range.
- **Regex Filtering**: `filterTransactionsByRegex` uses `java.util.regex` to perform case-insensitive matches on transaction descriptions.
- **Layered Application**: In `Parser.handleList` and `Parser.handleDelete` , these filters are applied sequentially (e.g., Filter by Date → Filter by Regex → Filter by Account).

3.8.2 Bulk Deletion Safety:

- To prevent accidental data loss, the `delete` command requires at least one filter flag (`-begin` , `-end` , `-match` , or `-acc`) when not deleting by a specific ID.
- The system calculates the list of matching transactions first and then iterates through the IDs to remove them from the master list.



3.8.3 Design Considerations

- **Predicate Composition:**
  - Filters are applied as a "stack."
  - **Decision:** We use Java Predicates to chain conditions. If a user provides `-acc Assets` and `-match Starbucks`, the system creates a composite predicate `(isAssets AND matchesStarbucks)`. This ensures that adding more filter types in the future does not require nested if-else blocks.
- **Non-Destructive Filtering:**

- The `list` command filters the view without removing items from the master `ArrayList`.

3.8.4 Alternatives Considered

- **Alternative 1: In-place Index Removal:** Iterating through the list and removing items that don't match.
  - **Pros:** Low memory overhead.
  - **Cons:** Very dangerous; a bug in the filter could wipe the user's data during a `list` operation.
  - **Decision:** The system always returns a *new* filtered list for display, keeping the master `TransactionsList` immutable during read operations.

3.9 Hierarchical Account Registry & Filtering Feature

Implementer: JJ

This feature allows users to filter transactions based on hierarchical account structures using the `list -acc` command.

3.9.1 Motivation

As the number of transactions grows, users need a way to quickly isolate transactions belonging to specific financial categories such as Assets, Expenses, or Income. Hierarchical account structures (e.g., `Assets:Bank:DBS`) make simple string matching insufficient. Hence, a structured filtering mechanism was implemented.

3.9.2 Implementation

This feature is implemented across three main components:

1. **Parser**
   - Detects the `-acc` flag
   - Extracts the account filter string
   - Passes it to `TransactionsList`
2. **Account**
   - Parses hierarchical account names using `:`
   - Provides `isUnder(String parentAccount)` method
   - Supports prefix-based hierarchical matching
3. **TransactionsList**
   - Iterates through all transactions
   - Filters postings based on account hierarchy

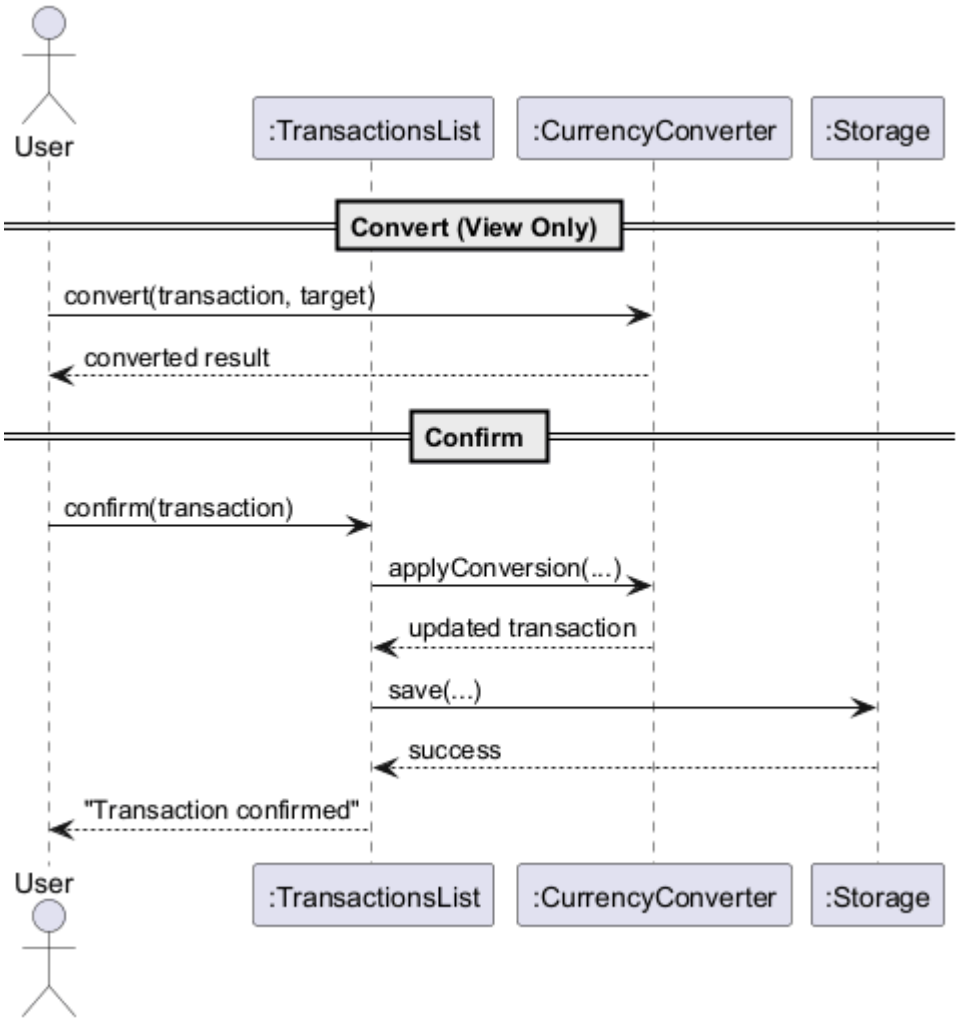
- Displays only matching transactions

3.9.3 Core Logic

The key method used is:

public boolean isUnder(String parentAccount)

3.9.4 Sequence Diagram



3.10 Quality Assurance and Logging Enhancements

Implementer: Chingy

This contribution focuses on improving code quality, maintainability, and debugging capabilities through comprehensive logging, error recovery, testing, and quality metrics.

3.10.1 Enhanced Logging System

Implemented a structured logging system to improve debugging and monitoring capabilities.

Key Components:

- **LoggingConfig**: Centralized logging configuration with JSON-formatted output
- **User Action Tracking**: Logs all user commands with timestamps and parameters
- **Structured Log Output**: JSON format for easy parsing and analysis
- **File-based Logging**: Logs stored in `logs/ledger67.log` for persistence

Implementation Details:

- **JSON Formatter**: Custom `JsonFormatter` class for structured log output
- **Log Levels**: Different log levels (INFO, WARNING, ERROR) for appropriate verbosity
- **User Action Method**: `userAction()` method for consistent user command logging
- **Automatic Setup**: Logging system initialized at application startup

Benefits:

- Improved debugging with structured log data
- Better monitoring of user interactions
- Easier troubleshooting of production issues
- Consistent log format across all components

3.10.2 Enhanced Error Recovery and Validation

Implemented comprehensive error recovery and validation to improve user experience and data integrity.

Key Components:

- **TransactionValidator**: Comprehensive validation with auto-correction suggestions
- **ValidationResult**: Structured validation results with warnings and suggestions
- **AutoCorrector**: Intelligent auto-correction for common input errors
- **Enhanced Error Messages**: User-friendly error messages with correction suggestions

Implementation Details:

- **Multi-stage Validation**: Validates date, description, currency, postings, and accounting equation
- **Auto-correction**: Suggests corrections for common errors (date format, currency codes, account names)
- **Validation Reports**: Detailed validation reports with actionable feedback
- **Common Issue Detection**: Detects and warns about common accounting issues

Benefits:

- Reduced user frustration with helpful error messages
- Improved data integrity through comprehensive validation
- Faster error resolution with auto-correction suggestions
- Better user guidance for complex accounting concepts

3.10.3 Enhanced Testing Framework

Implemented comprehensive testing including performance tests to ensure system reliability.

Key Components:

- **PerformanceTest**: Comprehensive performance testing suite
- **Quality Metrics**: Code coverage and quality checks
- **Build Integration**: Automated testing in build pipeline

Performance Test Coverage:

- **Transaction Filtering**: Tests filtering performance with large datasets (10,000+ transactions)
- **Balance Sheet Generation**: Tests balance sheet generation performance
- **Memory Usage**: Monitors memory consumption with large transaction volumes
- **Stress Testing**: Stress tests with concurrent operations
- **Validation Performance**: Tests transaction validation performance

Implementation Details:

- **Gradle Integration**: Performance tests integrated into build system
- **Separate Test Task**: `performanceTest` task for performance-specific tests
- **Realistic Data Generation**: Generates realistic test data for performance testing
- **Performance Metrics**: Measures execution time and memory usage

Benefits:

- Ensures system performance with large datasets
- Identifies performance bottlenecks early
- Provides confidence in system scalability
- Monitors resource usage patterns

3.10.4 Code Quality Metrics and Automation

Implemented comprehensive code quality metrics and automated quality checks.

Key Components:

- **Checkstyle**: Code style enforcement with lenient configuration
- **PMD**: Static code analysis for common programming flaws
- **SpotBugs**: Bug pattern detection
- **JaCoCo**: Code coverage analysis
- **Gradle Tasks**: Automated quality assurance tasks

Implementation Details:

- **Quality Assurance Task**: `qualityAssurance` task runs all quality checks
- **Comprehensive Build**: `comprehensiveBuild` task includes all tests and quality checks
- **Configuration Files**: Custom configurations for each quality tool
- **Coverage Requirements**: 45% minimum code coverage requirement

Quality Tools Configuration:

- **Checkstyle**: Lenient configuration for development, strict for production
- **PMD**: Custom ruleset focusing on critical issues
- **SpotBugs**: Exclusion filters for false positives
- **JaCoCo**: Coverage verification with reasonable thresholds

Benefits:

- Consistent code quality across the codebase
- Early detection of potential issues
- Automated quality enforcement
- Measurable quality metrics

3.10.5 Build System Enhancements

Enhanced the Gradle build system for better development workflow.

Key Enhancements:

- **Quality Assurance Pipeline**: Integrated quality checks into build process
- **Performance Testing**: Separate performance test execution
- **Coverage Reporting**: Automated coverage reports
- **Task Organization**: Well-organized Gradle tasks for different purposes

Gradle Tasks:

- `qualityAssurance` : Runs all code quality checks
- `comprehensiveBuild` : Full build including tests and quality checks
- `performanceTest` : Runs performance tests separately

- `jacocoTestReport` : Generates code coverage reports

Benefits:

- Streamlined development workflow
- Automated quality enforcement
- Comprehensive testing pipeline
- Better developer experience

3.10.6 Design Considerations

- **Progressive Enhancement**: Quality tools configured with reasonable defaults
- **Developer Experience**: Focus on helpful feedback rather than blocking development
- **Performance Focus**: Emphasis on system performance and scalability
- **Maintainability**: Code structured for easy maintenance and extension

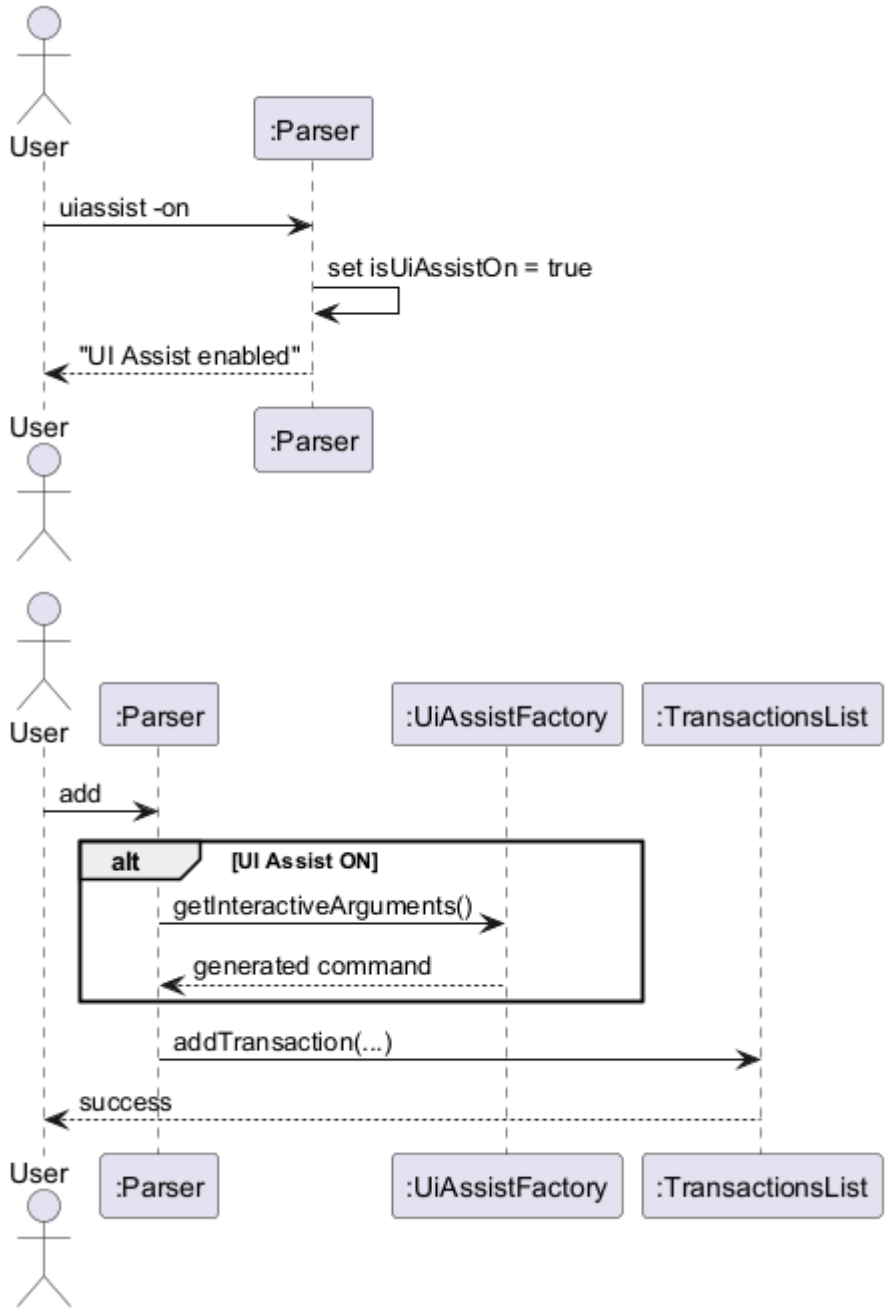
3.10.7 Impact on Development Process

- **Improved Code Quality**: Consistent code style and fewer bugs
- **Better Debugging**: Enhanced logging for easier troubleshooting
- **Performance Confidence**: Performance testing ensures system scalability
- **Automated Quality**: Quality checks integrated into development workflow
- **Comprehensive Testing**: Both functional and performance testing coverage

3.11 UI Improvements and Assistance

Implementer: Pran

To bridge the gap between a powerful CLI and a user-friendly experience, Ledger67 implements a "UI Assistance" mode. This feature caters to users who find long command-line flags (like `-date` , `-desc` , `-p`) difficult to remember.



3.11.1 Implementation Details

The `uiassist` feature is implemented as a state-toggle within the `Parser`.

- **State Management:** When `uiassist -on` is executed, the `Parser` sets an internal boolean `isUiAssistEnabled`.
- **Prompt-Driven Input:** Instead of expecting a single line of input, the `Parser` initiates a loop that prompts the user for individual fields:
  i. "Enter Date (DD/MM/YYYY):"
  ii. "Enter Description:"
  iii. "Enter Currency (SGD/USD/EUR):"
  iv. "Enter Postings (e.g., 'Assets:Cash -50'). Type 'done' to finish."

3.11.2 Design Considerations

- **Consistency:** The UI Assist mode uses the exact same validation logic as the manual mode. Whether a user uses the flags or the prompts, the same `DateValidator` and `BalanceChecker` are invoked.
- **Non-Blocking Toggles:** Users can toggle UI Assist off at any time. This allows power users to use flags for quick entries and UI Assist for complex transactions.

3.11.3 Alternatives Considered

- **Alternative 1: Interactive CLI Menus (Arrow Keys):** Using a library like JLine to create selectable menus.
  - **Pros:** Very "modern" feel.
  - **Cons:** Heavy dependency on external libraries and may not work consistently across all terminal emulators (CMD vs. Bash).
  - **Decision:** Standard `System.out` prompts were used to ensure 100% compatibility across all operating systems and environments.
- **Alternative 2: Forced Guided Mode:** Making the app guided-only for all `add` commands.
  - **Pros:** No need to learn flags.
  - **Cons:** Frustrating for experienced users who can type a single-line command faster than answering four prompts.
  - **Decision:** The optional toggle `uiassist -on/-off` provides the best of both worlds.

3.12 Documentation and User Experience Improvements

Implementer: Chingy

This contribution focuses on enhancing the user experience through comprehensive documentation and implementing the help system to make the application more accessible to new users.

3.12.1 User Guide Rewrite and Enhancement

The User Guide was completely rewritten to provide comprehensive documentation for both new and experienced users.

Key Improvements:

- **Double-Entry Accounting Explanation**: Added beginner-friendly explanations of double-entry bookkeeping concepts including:
  - The accounting equation (Assets = Liabilities + Equity)
  - Debit vs. Credit fundamentals
  - How transactions work in Ledger67
  - Practical examples for different user scenarios

- **Currency Conversion Documentation**: Added detailed documentation for all currency-related features:
  - `convert` command for currency conversion
  - `convert transaction` command for converting existing transactions
  - `rates refresh` command for updating exchange rates
  - Complete examples and expected outputs
- **Command Reference Enhancement**: Updated the command summary table to include all new commands with clear formatting and examples.
- **FAQ Section**: Added frequently asked questions to address common user concerns about:
  - Data transfer between computers
  - Difference between debit and credit
  - Personal finance usage
  - Supported currencies
  - Double-entry system implementation

3.12.2 Help Command Implementation

Implemented a comprehensive help system within the application to provide immediate assistance to users.

Implementation Details:

- **Parser Integration**: Added `handleHelp()` method to the `Parser` class
- **Command Structure**: Organized help output into logical sections:
  - Available commands with clear numbering
  - Format specifications for each command
  - Practical examples for each command
  - Additional information about data formats and constraints
- **User-Friendly Output**: The help command displays:
  - All 10 available commands with consistent formatting
  - Required and optional parameters for each command
  - Real-world usage examples
  - Important notes about data storage and exchange rates

Technical Implementation:

```java
private void handleHelp() {
    System.out.println("=== Ledger67 Help ===");
    System.out.println("Available commands:");
```

```
    // ... command documentation
}
```

3.12.3 Documentation Maintenance and Currency Feature Updates

Ensured documentation stays current with evolving project features.

Accomplishments:

- **Real-time Updates**: Updated User Guide immediately after currency conversion features were merged
- **Command Consistency**: Verified help command output matches actual implemented commands
- **Build Integration**: Tested help functionality across different execution methods (Gradle run, JAR file)
- **Team Collaboration**: Committed and pushed documentation updates to ensure team synchronization

3.12.4 Design Considerations

- **Beginner-Friendly Approach**: Documentation written for users with no prior accounting knowledge
- **Progressive Disclosure**: Basic concepts explained first, followed by advanced features
- **Consistent Formatting**: All commands documented with the same structure for predictability
- **Practical Examples**: Real-world scenarios provided for each command
- **Accessibility**: Help available both in-app and via external documentation

3.12.5 Impact on User Experience

- **Reduced Learning Curve**: New users can understand the application faster
- **Immediate Assistance**: Users can get help without leaving the application
- **Comprehensive Reference**: All features are properly documented
- **Consistent Experience**: Documentation matches actual application behavior

4. Appendix A: Product Scope

4.1 Target User Profile

Ledger67 is designed for:

- Individuals who want to track personal financial transactions
- Small business owners who require simple expense tracking
- Students learning about financial management and accounting
- Users comfortable with command-line interfaces
- Users who prefer typing over graphical interfaces
- Users dealing with multiple currencies who require quick and reliable currency conversion

4.2 Value Proposition

Ledger67 addresses several key needs:

1. **Simplified Financial Tracking**
   Provides a structured way to record and manage financial transactions using double-entry bookkeeping.

2. **Rapid Data Entry**
   Command-line interface enables faster input for users familiar with typing workflows.

3. **Portability**
   Lightweight Java application that runs on any system with Java 17+.

4. **Multi-Currency Support**
   Allows users to convert and view transactions across currencies without modifying stored data.

5. **Live Exchange Rate Integration**
   Supports real-time rate updates for more accurate financial comparisons.

5. Appendix B: User Stories

Version	As a ...	I want to ...	So that I can ...
v1.0	new user	see usage instructions	refer to them when I forget how to use the application
v1.0	user	add a new transaction	record my financial activities
v1.0	user	list all transactions	view my transaction history

Version	As a ...	I want to ...	So that I can ...
v1.0	user	edit a transaction	correct mistakes in previously recorded transactions
v1.0	user	delete a transaction	remove erroneous or duplicate entries
v1.0	user	have my transactions automatically saved	avoid losing data between sessions
v1.0	user	load previously saved transactions	continue tracking finances across sessions
v1.0	user	convert currencies	understand values across different currencies
v1.0	user	convert a transaction to another currency	quickly view equivalent values without modifying stored data
v1.0	user	view transactions in another currency	compare spending consistently across currencies
v1.0	user	refresh exchange rates	ensure conversion uses up-to-date rates
v2.0	user	filter transactions by date	review transactions from specific time periods
v2.0	user	search transactions by description	find specific transactions quickly
v2.0	user	validate my transactions	ensure my double-entry accounts are balanced
v2.0	user	add new categories under each account type	better organize financial data
v2.0	user	categorise transactions	filter by transaction type
v2.0	user	generate balance sheet	understand financial position
v2.0	user	export balance sheet	analyse data externally
v2.0	user	filter balance sheet by account	focus on specific categories

6. Appendix C: Non-Functional Requirements

6.1 Reliability

- The application should not crash on invalid user input
- Data should remain consistent during all CRUD operations
- Error messages should be informative and actionable
- Exchange rate data should be validated before use

6.2 Usability

- Command syntax should be intuitive and consistent
- Help text should be accessible via a help command
- Error messages should clearly indicate what went wrong and how to fix it
- Conversion commands should clearly distinguish between stored data and display-only values

6.3 Maintainability

- Code should follow Java coding standards
- Critical functionality should be covered by unit tests
- Code should include meaningful documentation and comments
- Separation of concerns should be maintained across components

7. Appendix D: Glossary

- **Transaction**: A financial record containing a date, description, currency, and a list of postings that together form a balanced entry.
- **Posting**: A single entry linking an account to a numerical amount within a transaction.
- **Account**: A category used to classify financial data (e.g., Assets, Expenses, Income), supporting hierarchical structures.
- **Double-Entry Accounting**: An accounting method where every transaction affects at least two accounts and must balance.
- **Currency Code**: Three-letter ISO currency code (SGD, USD, EUR).
- **Parser**: Component responsible for interpreting user commands and delegating execution.
- **TransactionsList**: Component managing all transaction objects and performing filtering, updates, and aggregation.
- **Validation**: Ensuring that user inputs meet required constraints before processing.
- **Auto-increment ID**: Unique identifier automatically assigned to each transaction.

- **CRUD Operations**: Create, Read, Update, Delete operations for managing data.
- **Balance Sheet**: A financial statement summarizing Assets, Liabilities, and Equity.

8. Appendix E: Instructions for Manual Testing

This section provides guidance for testers to validate the key features of Ledger67.
It complements the User Guide by outlining important test flows and edge cases.

8.1 Launching the Application

Preconditions:

- Java 17 or above is installed
- Application is compiled or JAR file is available

Steps:

1. Navigate to the project directory
2. Run:

```
java -jar tp.jar
```

or

```
./gradlew run
```

8.2 Adding Transactions

Test basic add:

```
add -date 01/01/2026 -desc "Lunch" -p "Assets:Cash -10" -p "Expenses:Food 10" -c SGD
```

Expected:

- Transaction is added successfully
- `list` shows the new transaction

8.3 Validation of Minimum Postings

Test invalid case:

```
add -date 01/01/2026 -desc "Invalid" -p "Assets:Cash 10" -c SGD
```

Expected:

- Error indicating at least 2 postings required
- Transaction is NOT added

8.4 Listing and Filtering

```
list
list -acc Expenses
list -match Lunch
list -begin 01/01/2026 -end 31/12/2026
```

Expected:

- Filters are applied correctly
- No data is modified

8.5 Editing Transactions

Valid edit:

```
edit 1 -desc "Updated Lunch"
```

Invalid edit (1 posting):

```
edit 1 -p "Assets:Cash -20"
```

Expected:

- Valid edit succeeds
- Invalid edit is rejected
- Original transaction remains unchanged

8.6 Deleting Transactions

```
delete 1
delete -match Lunch
```

Expected:

- Correct transactions are removed
- Bulk delete requires at least one filter

8.7 Currency Conversion

```
convert -a 100 -from USD -to SGD
convert transaction 1 -to SGD
```

Expected:

- Conversion is displayed
- Stored values remain unchanged until confirmed

8.8 Confirm Feature

```
convert transaction 1 -to SGD
confirm
```

Expected:

- Transaction is updated permanently
- Currency and values are changed

8.9 List Conversion (View Only)

```
list -to USD
```

Expected:

- Converted values displayed
- Original data remains unchanged

8.10 Balance Sheet

```
balance
balance -acc Assets
balance -to USD
```

Expected:

- Aggregated totals displayed
- Equation check shown
- Filtered view may not balance

8.11 Monetary Precision (Rounding)

```
add -date 01/01/2026 -desc "Precision Test" -p "Assets:Cash 100.999999" -p "Expenses:Food
```

Expected:

- Values are stored as 2 decimal places
- Example: 100.999999 → 101.00

8.12 UI Assist Mode

```
uiassist -on
add
```

Expected:

- Prompts guide input step-by-step
- Resulting transaction is valid

8.13 Edge Cases

Test the following:

- Invalid date format
- Invalid currency
- Unbalanced transactions

- Duplicate flags
- Empty input

Expected:

- System rejects invalid input gracefully
- No crash occurs

8.14 Notes for Testers

- Stored data is persisted automatically after each modification
- Conversion features are **non-destructive** unless confirmed
- Filtering operations do not modify stored data